

Project 2 Report – Systems Biology

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1 Part A

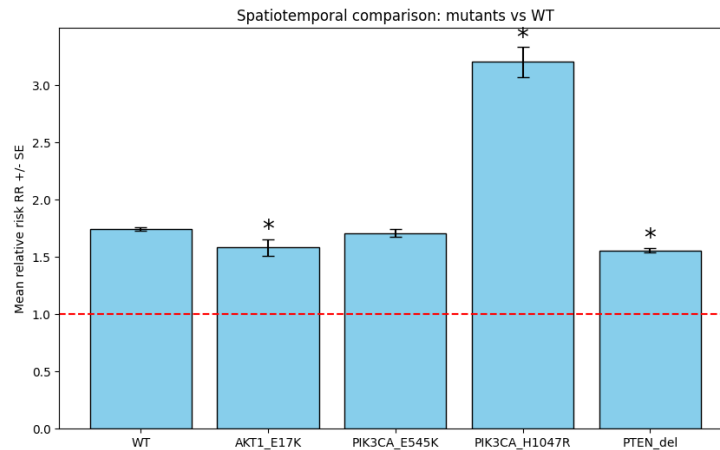
1.1 Task A1

1.1.1 Brief methods summary

The analysis of the effect of mutations on ERK signal dynamics was documented in the `TaskA1_MutationComparison.ipynb` notebook, which uses the `compare_spatiotemporal_behavior.py` file from the `spatiotemporal-continuation/scripts/` directory. The analysis included five cell lines: wild type (WT) and the following mutants: `AKT1_E17K`, `PIK3CA_E545K`, `PIK3CA_H1047R`, and `PTEN_del`. Standard analysis parameters were used: spatial radius (`spatial_radius`) of 60, time window (`future_window_frames`) of 3, and signal column `ERKKTR_ratio`. Mean relative risk values and statistical significance tests were calculated based on data from the `block_level_summary.csv` file, allowing comparison of value distributions between mutation groups and the wild type (WT). Mean relative risk data (`mean_relative_risk`) were extracted from the `group_level_summary.csv` file. To verify the statistical significance of differences between WT and individual mutants, Mann–Whitney U tests with Bonferroni correction for multiple comparisons were performed.

1.1.2 Key results

The main result of the analysis is a comparative table and a bar chart illustrating the differences in signal spread.



	mutation	mean_RR	std_error	p_value_vs_WT	significant
0	WT	1.742349	0.017193	NaN	False
1	AKT1_E17K	1.581421	0.071932	1.384587e-02	True
2	PIK3CA_E545K	1.707110	0.031885	7.341158e-01	False
3	PIK3CA_H1047R	3.200310	0.130643	1.225466e-08	True
4	PTEN_del	1.556633	0.021624	1.464234e-06	True

1.1.3 Interpretation

The red dashed line in the chart ($RR = 1$) represents the absence of spatiotemporal correlation. Values above one indicate the existence of a signal transmission mechanism within the cell population. Statistical analysis revealed that the PIK3CA_H1047R mutation showed a relative risk value substantially higher than both the red reference line and the WT group. The remaining mutants also exhibited values above the red line, but they were similar to or lower than the WT level. This suggests that these mutations may paradoxically make signal propagation more diffuse than in the wild type.

A key finding of the analysis was the dramatic increase in relative risk observed for the PIK3CA_H1047R mutation ($RR \approx 3.20$), indicating exceptionally strong signaling coupling between neighboring cells. The PIK3CA_H1047R mutation is located within the kinase domain of the PI3K protein, leading to constitutive and strong pathway activation. As a consequence, cells become highly sensitive to signals originating from neighboring cells. Activation of one cell may therefore trigger a rapid chain reaction in adjacent cells, substantially increasing both the range and intensity of signaling across the population. In contrast, the remaining mutations may induce weaker but chronic pathway activation, potentially leading to cellular habituation to signaling stimuli. As a result, cells respond less efficiently to neighboring-cell activation, making signal transmission less efficient and more chaotic.

1.2 Task A2

1.2.1 Research question

Do different PI3K–AKT pathway mutations exhibit different spatiotemporal relay timescales for ERK signal propagation? Specifically, does the lag τ^* at which the Relative Risk $RR(\tau)$ is maximised differ between wild-type cells and cells carrying oncogenic mutations?

1.2.2 Methods

For each cell–frame observation (cell c , frame t) we define:

- **Exposure:** at least one spatial neighbour of cell c shows a *jump event* (a large upward signal change) at frame t .
- **Response at lag τ :** cell c itself shows a jump event at frame $t + \tau$.

A *jump event* is defined as a frame-to-frame signal increase $\Delta s \geq \theta$, where θ is the 90th percentile of all positive signal differences within the block (data-driven, per-block threshold). Spatial neighbours are cells within a Euclidean radius of 60 image-coordinate units at the same frame, identified via a k-d tree. The Relative Risk at lag τ is:

$$\text{RR}(\tau) = \frac{P(\text{focal cell jumps at } t + \tau \mid \text{neighbour jumped at } t)}{P(\text{focal cell jumps at } t + \tau \mid \text{no neighbour jumped at } t)} \quad (1)$$

We evaluated $\tau \in \{0, 1, 2, 3, 4, 5, 6\}$ frames (0, 5, 10, 15, 20, 25, 30 min at 5 min/frame acquisition). For each lag and mutation, raw counts were pooled across all experiment-site blocks before computing $\text{RR}(\tau)$, avoiding bias from averaging ratios across blocks of different sizes. Signal: `ERKKTR_ratio`.

Three cell lines were selected: **WT** (baseline), **AKT1_E17K** (gain-of-function point mutation constitutively activating AKT1 independently of PI3K), and **PTEN_del** (loss of PTEN phosphatase, leading to chronically elevated PIP3 and secondary AKT hyperactivation). Both mutants converge on AKT hyperactivation through mechanistically distinct nodes, making them informative for comparing ERK relay kinetics. Blocks analysed: WT $n = 24$, AKT1_E17K $n = 21$, PTEN_del $n = 27$.

1.2.3 Key results

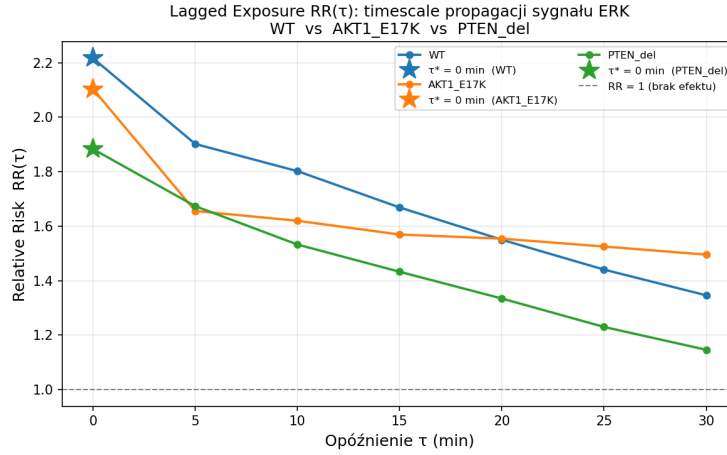


Figure 1: Lagged Exposure $\text{RR}(\tau)$ vs. lag τ (min) for WT (blue), AKT1_E17K (orange), and PTEN_del (green). Stars mark τ^* ; dashed line: $\text{RR} = 1$.

Table 1: Optimal lag τ^* and maximum Relative Risk for each cell line.

Mutation	τ^* (frames)	τ^* (min)	max RR
WT	0	0	2.216
AKT1_E17K	0	0	2.103
PTEN_del	0	0	1.883

1.2.4 Interpretation

The lagged exposure analysis reveals that for all three genotypes the maximum Relative Risk occurs at $\tau^* = 0$ frames (0 min): WT achieves $RR(0) = 2.22$, AKT1_E17K reaches $RR(0) = 2.10$, and PTEN_del attains $RR(0) = 1.88$. The absence of a detectable positive lag is consistent with two scenarios that cannot be distinguished at the available temporal resolution of 5 min/frame: either ERK signal propagation between neighbours occurs on a sub-5-minute timescale, or neighbouring cells respond nearly simultaneously to a shared paracrine stimulus rather than relaying the signal sequentially. In all three genotypes $RR(\tau)$ decreases monotonically with increasing lag, confirming that spatial coordination of ERK jumps is a short-lived phenomenon.

A clear gradient in the magnitude of $RR(0)$ is observed: $WT > AKT1_E17K > PTEN_del$. Oncogenic mutations hyperactivating the PI3K–AKT axis therefore reduce, rather than amplify, the spatial coordination of ERK jumps. A plausible explanation is that constitutive AKT activity elevates the rate of autonomous ERK jumps independent of neighbour signalling, increasing the baseline jump rate in unexposed cells and diluting the relative contribution of neighbour-driven activation. PTEN_del, acting upstream of AKT through chronic PIP3 elevation, produces the strongest tonic activation and hence the largest autonomous noise, yielding the lowest RR. In summary, whereas $\tau^* = 0$ for all lines precludes distinguishing propagation timescales at this frame rate, the analysis uncovers a biologically meaningful difference in the *strength* of spatial coordination: oncogenic PI3K–AKT mutations attenuate the synchronised propagation of ERK activation waves across the cell population.

1.3 Task A3

1.3.1 Research question

Are the mutation-level differences in ERK spatiotemporal coupling (relative risk) stable across reasonable choices of the three main analysis parameters?

1.3.2 Methods

A one-at-a-time (OAT) sensitivity analysis was performed, varying each of the three principal parameters while holding the other two at their baseline values ($r = 60$, $W = 3$, $q = 0.90$). The ranges evaluated were:

- **Spatial radius** $r \in \{30, 60, 90, 150\}$ image-coordinate units (neighbourhood size);
- **Future window** $W \in \{1, 3, 6, 12\}$ frames (5, 15, 30, 60 min);
- **Jump quantile** $q \in \{0.70, 0.80, 0.90, 0.95\}$ (threshold stringency).

All five genotypes (WT, AKT1_E17K, PIK3CA_E545K, PIK3CA_H1047R, PTEN_del) were included. To keep runtime manageable, the analysis used Experiment 1 only (19 experiment-site blocks; one WT block was absent from the dataset). For each parameter setting, raw counts were pooled across blocks before computing the Relative Risk, consistent with the approach in Tasks A1 and A2.

1.3.3 Key results

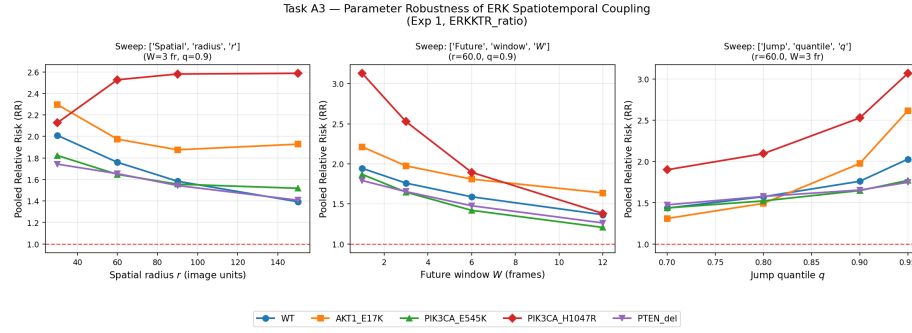


Figure 2: Pooled Relative Risk per mutation as a function of each swept parameter (Exp 1, ERKKTR_ratio). Dashed red line: $RR = 1$. Left: spatial radius sweep ($W = 3$ fr, $q = 0.90$). Centre: future-window sweep ($r = 60$, $q = 0.90$). Right: jump-quantile sweep ($r = 60$, $W = 3$ fr).

1.3.4 Interpretation

The OAT analysis shows that the elevated ERK spatiotemporal coupling of PIK3CA_H1047R is broadly robust, but two rank inversions occur at extreme parameter values.

Spatial radius. At the baseline and larger radii ($r \in \{60, 90, 150\}$), PIK3CA_H1047R consistently achieves the highest pooled RR (2.53, 2.58, 2.59 respectively). As r increases from 60 to 150, absolute RR values decline modestly for all mutations except PIK3CA_H1047R and AKT1_E17K, whose gap from the rest widens. At the smallest radius ($r = 30$), however, a rank inversion occurs: AKT1_E17K ($RR = 2.30$) overtakes PIK3CA_H1047R ($RR = 2.13$). This may reflect the fact that a 30-unit neighbourhood captures only immediately adjacent cells, where AKT1_E17K-driven contact-dependent signalling may be particularly prominent.

Future window. PIK3CA_H1047R shows the strongest coupling for short and intermediate windows: $RR = 3.13$ at $W = 1$ frame (5 min), 2.53 at $W = 3$ (15 min), and 1.89 at $W = 6$ (30 min). At $W = 12$ frames (60 min), all mutations converge toward $RR \approx 1$, and a second inversion occurs: AKT1_E17K (1.64) surpasses PIK3CA_H1047R (1.38). This pattern is consistent with the interpretation that PIK3CA_H1047R mediates fast, tightly coupled ERK propagation that becomes diluted when many autonomous jumps accumulate over the 60-minute window.

Jump quantile. PIK3CA_H1047R retains the highest RR across all four quantile levels (1.90 at $q = 0.70$; 2.09 at $q = 0.80$; 2.53 at $q = 0.90$; 3.07 at $q = 0.95$), with no rank inversions. The ordering within the remaining mutations shifts at extreme thresholds: AKT1_E17K is the weakest at $q = 0.70$ ($RR = 1.31$) but second-strongest at $q = 0.95$ ($RR = 2.62$), suggesting that AKT1_E17K’s spatial coupling is preferentially visible for the sharpest, most unambiguous jump events.

Conclusion. The finding that PIK3CA_H1047R exhibits exceptionally strong ERK spatiotemporal coupling is largely robust: it holds for all baseline and larger spatial radii, for short and intermediate future windows, and across all jump thresholds. The two rank inversions (both favouring AKT1_E17K) arise only at extreme parameter values—the very smallest neighbourhood ($r = 30$) and the very largest temporal window ($W = 12$)—and are biologically interpretable rather than indicative of an artefact.

1.4 Task B

We chose option 2 - open research question.

1.4.1 Introduction

Research question: **Does the size of a group of cells affect how effectively they transmit a signal to each other?**

Does a signal travel more easily in a large group of cells or in a small group? In small groups of cells, the signal can quickly decay because once the wave reaches the vessel edge, it has nowhere else to go. In contrast, in large cell clusters, the signal can circulate and self-propel itself, creating a chain reaction. Determining whether the relative risk (RR) depends on the number of cells (n_nodes) will help answer the question of whether cells need to reach a sufficiently large tissue size for efficient communication.

Hypothesis: **The greater the total number of cells in the study group, the higher the relative risk (RR). Large cell populations provide more stable conditions for signal propagation.**

1.4.2 Methods

The analysis was performed on data from the `block_level_summary.csv` file. The variable `n_nodes` (the total number of nodes in the spatiotemporal graph

for a given block) was used as a measure of network size. The dependent variable was relative risk (relative_risk).

To test whether population size affects communication, the Spearman rank correlation coefficient (ρ) between n_nodes and relative_risk was calculated. The analysis was performed for the control group (WT) and selected mutants to test whether genetic mutations make cells independent of the tissue-scale effect.

1.4.3 Results

The analysis of the relationship between network size and signal propagation is based on two visualizations: a plot of the population size distribution (violin/boxplot) and a plot of the dependence of the relative risk on the total number of nodes (scatterplot with trend lines). Here are the main conclusions:

- Diversity of network sizes



Figure 3: The violin plot reveals that individual cell lines naturally form networks of varying sizes — observe the height of the widest parts of the plot. Cells with the PIK3CA_H1047R and PTEN_del mutations form the smallest populations (typically below 250,000 nodes). In contrast, the AKT1_E17K and PIK3CA_E545K mutations dominate populations with much larger sizes (over 300,000 nodes). Wild type (WT) falls in the middle.

- The dependence of RR on network size is mutation-specific

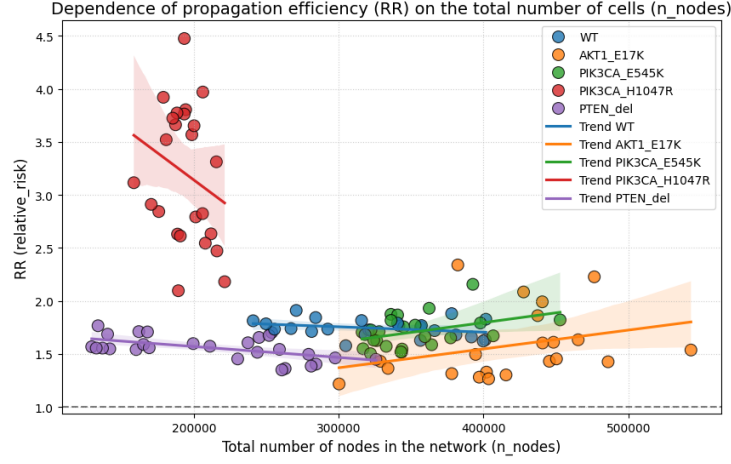


Figure 4: In the case of the control group (WT) and *PTEN_{del}* mutants, the trend lines are almost completely flat, indicating a lack of correlation between the population size and the RR value. For the *AKT1_E17K* and *PIK3CA_E545K* mutations, a clear positive correlation was observed — the propagation efficiency (RR) increases with the increase in the number of cells in the network. The completely opposite and strongest effect was observed for *PIK3CA_H1047R*. This mutation shows a negative correlation — it reaches extremely high RR values in the smallest populations, and with the increase in network size, its efficiency decreases.

1.4.4 Discussion

The obtained results refute the hypothesis of a universal, positive effect of population size on signal propagation, revealing substantial differences between mutations. Cells with the *PIK3CA_H1047R* mutation are naturally so active that they do not require a large number of neighboring cells. In small groups, their signal is highly concentrated and triggers a strong chain reaction.

The *AKT1_E17K* and *PIK3CA_E545K* mutations show the opposite behavior. Individual cells emit a relatively weak signal, and therefore require the support of a large population for effective communication. In contrast, healthy cells (WT) and *PTEN_{del}* mutants display stable communication efficiency. This suggests that their internal signaling systems are well-regulated and capable of maintaining signal transduction independently of tissue size.

The main limitation of this analysis is that the studied mutations naturally generate populations of very different sizes. For example, *PIK3CA_H1047R* cells occurred mainly in small groups, whereas *AKT1_E17K* cells appeared predominantly in large populations. This lack of overlap across the graph axes makes it difficult to determine whether the observed trends in RR are caused by the number of cells itself or simply reflect intrinsic properties of a given cell line.

To address this limitation, future experiments should artificially control culture size, for example by forcing each mutation to grow in populations of identical size. Only such an experimental design would allow us to determine whether population size exerts a universal effect on signal propagation independently of mutation type.